

Technical Report: Secure Offline Mesh Network Architecture

1. Executive Summary

This report details the architecture, hardware integration, and software implementation of a secure, offline IoT messaging system utilizing ESP32 microcontrollers. Designed to bypass traditional router-based infrastructure, the system leverages the ESP-NOW protocol to establish a low-latency, peer-to-peer mesh network. The physical interface integrates 4x4 matrix keypads, 16x02 I2C Liquid Crystal Displays, and piezoelectric buzzers, forming fully autonomous communication nodes. By combining C++ inter-node messaging logic with a lightweight XOR encryption cipher and Bluetooth (BLE) gateway capabilities, the system provides a robust framework for secure localized signalling and emergency alert broadcasting.

2. System Architecture and Methodology

The core communication architecture relies on ESP-NOW, a connectionless Wi-Fi communication protocol that allows direct frame transmission between registered MAC addresses without requiring a centralized access point. Each ESP32 unit is configured in Wi-Fi Station (WIFI_STA) mode. Node A acts as the primary transmission interface, scanning the 4x4 keypad matrix for user input. Upon detecting a keypress, the embedded logic maps the input to a predefined string (e.g., 1 translates to "Hello", * translates to "Alert"). The payload is then encrypted and broadcasted to the designated receiver. Node B operates as the receiver, decrypting incoming payloads, rendering the text on the I2C LCD, and triggering an auditory alert via the piezoelectric buzzer.

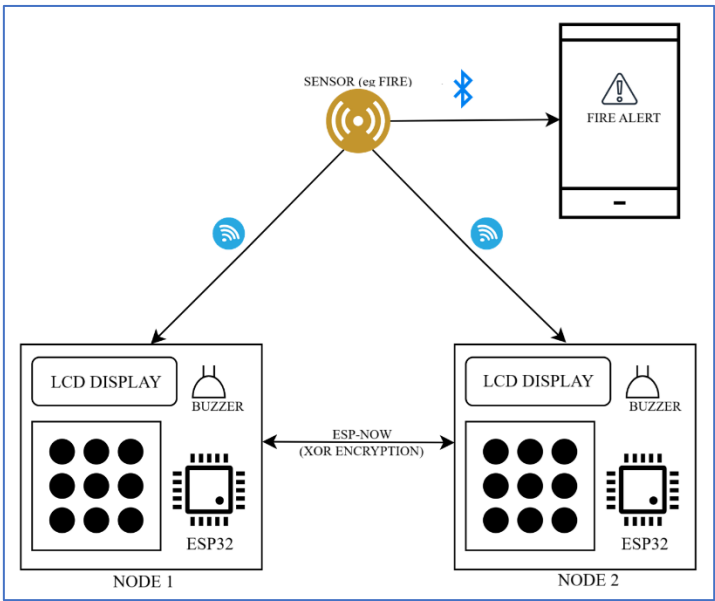


Figure 1: Hardware schematic detailing the integration of the ESP32 with the matrix keypad, I2C LCD, and buzzer.

3. Hardware Integration

The hardware topology was designed for localized, 5V battery-powered operation. The component integration is defined as follows:

- **Microcontroller:** ESP32-WROOM-32 Development Modules process all I/O polling, cryptographic operations, and RF transmissions.
- **Input Interface:** A standard 4x4 membrane keypad utilizes 8 GPIO pins (Rows: 19, 18, 32, 33; Cols: 25, 26, 27, 14) to capture user commands.
- **Output Interface:** A 16x02 LCD interfaces via the I2C bus (SDA: GPIO 21, SCL: GPIO 22) for minimal pin consumption, while a passive buzzer is driven by a 1kHz PWM signal on GPIO 13 to provide immediate auditory feedback upon payload reception.

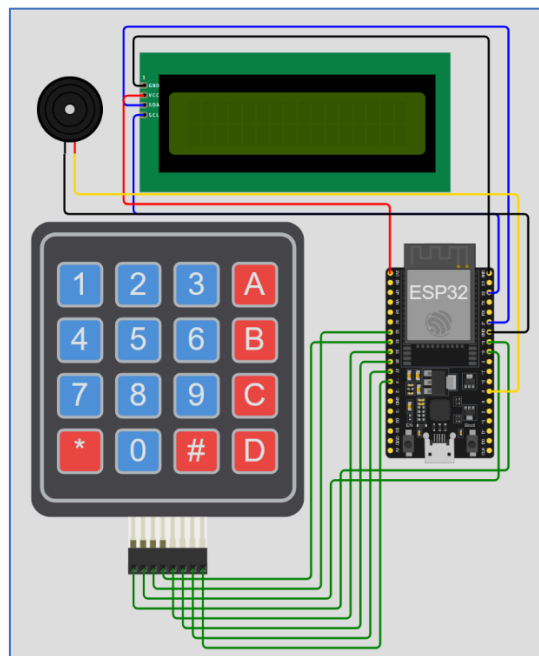


Figure 2: System topology and logical data flow between nodes.

4. Software Implementation and Cryptography

The system operates on a C++ codebase utilizing asynchronous callback functions (`esp_now_register_send_cb` and `esp_now_register_rcv_cb`) to handle network traffic without blocking the main execution loop.

To secure the transmitted data against localized packet sniffing, a symmetric XOR cipher was implemented. Before transmission, the plain-text payload is iterated through a byte-wise XOR operation against a static hexadecimal key (0xAA). The receiving node applies the identical XOR operation to the buffered bytes to reconstruct the original string.

Additionally, a BluetoothSerial instance is initialized to act as an external gateway. If a critical "Alert" payload is transmitted across the mesh, the receiving ESP32 forwards an emergency notification string over BLE to any paired smartphone, bridging the localized offline mesh with personal mobile devices.

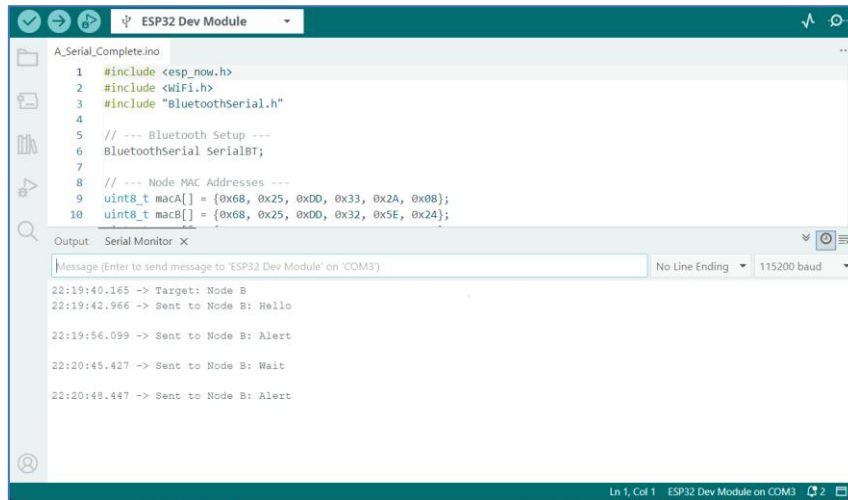


Figure 3: Mobile device intercepting the emergency alert via the Bluetooth gateway.

5. Technical Constraints and Coexistence

During system validation, a hardware coexistence constraint was identified within the ESP32's RF architecture: simultaneous utilization of the 2.4 GHz radio for both ESP-NOW (Wi-Fi) reception and active Bluetooth (BLE) connections can result in dropped frames. To mitigate this interference, BLE is primarily designated to the transmitting node for outbound alert forwarding, while the receiving node dedicates its radio purely to the ESP-NOW protocol to ensure uninterrupted message reception.

6. Conclusion

The completed hardware prototype successfully validates the viability of offline, encrypted mesh networks for critical localized communication. By synthesizing ESP-NOW protocols, robust cryptographic byte manipulation, and responsive hardware peripherals, the architecture presents a highly scalable foundation for industrial signalling, secure facility messaging, and remote IoT emergency alert systems.